Aditya Pandya

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Education

University of Waterloo Sep. 2023 – Apr. 2028 (Expected)
BASc Computer Engineering, Co-operative Program *Waterloo, ON*
- **Coursework:** Data Structures and Algorithms, Database Systems, Concurrency, Probability and Statistics

Work Experience

Member of Technical Staff Intern Jan. 2026 – Apr. 2026
Customer Maps *Toronto, ON*
- Shipped a production **multi-agent** system: a **Gemini 2.5 Flash** root agent planning over **16 function tools** and delegating to **3 sub-agents** (sandboxed Python, dashboards, reporting), streaming tool calls to a **Next.js** client over **SSE**.
- Built **4 MCP servers** (Shopify, GA4, Google Drive, Slack) exposing **32 tools** behind per-user credential and context injection; indexed **OCR'd documents** in SQLite **FTS5** so agents could retrieve from messy PDFs instead of bespoke clients.
- Locked **33 analysis tools** behind **99 pytest contract evals** (schema, types, failure modes) and held staging inside a **3s** budget on **50K+** row pulls via in-memory frame handles and a TTL spill cache.
- Shipped a **14KB** storefront pixel batching signals on **2s/20-event** windows into per-shop Pub/Sub topics, merged into Firestore live state and wide BigQuery rows feeding model features.

Full Stack Developer May 2025 – Aug. 2025
Crbit Inc. *Toronto, ON*
- Shipped a **GPT-4 tool-calling** financial assistant over transaction history; **grounded** every numeric claim in per-user SQL aggregates (no model arithmetic), **cutting support queries 18%**.
- Built Python **FastAPI** ingestion APIs normalizing schemas across banks, **cutting retrieval latency 35%** via async fan-out and cached provider responses.
- Streamed balances to a **React Native** dashboard for **1,000+** concurrent users (**+22%** session duration); OAuth2/JWT on every LLM and banking endpoint.

Machine Learning Engineer Intern Aug. 2024 – Dec. 2024
Andromeda360 *Austin, TX*
- Built a **LangChain** + **FastAPI RAG** service over **Elasticsearch** for GPT-4 conversational analytics, **cutting query latency 35%** and raising throughput **40%**.
- Designed table-augmented generation in **Haystack**: transformer semantic search plus structured table retrieval. **Eval: +50% retrieval precision**, **+30% grounded-answer accuracy**.
- Containerized training and serving with **Docker**, cutting **training time 20%** and lifting **accuracy 25%**.

Software Developer Dec. 2023 – May 2024
Counting Opinions *Toronto, ON*
- Built a GPT-3 NL-to-visualization chatbot in **React** over **1M+** rows (**+20%** query efficiency) and a schema-validated JSON pipeline with a **RAG** retrieval stage for grounded NLU.

Projects

Multi-Agent Synthetic Market Simulator | *Google ADK, Gemini 2.5 Pro/Flash, Pydantic, asyncio, Vertex AI*
- Built an async **6-agent** pipeline (persona → strategy → 12-month ops loop → classifier → aggregator) simulating **30 businesses** over **360 agent-months** and surfacing **221** distinct friction patterns.
- Routed **Flash** for structured generation and **Pro** for synthesis; saturation early-stop (**<5% new tags**/batch) and JSON-repair retries as a cheap **eval**/cost loop instead of full re-prompts.

Personalization Inference Service | *ONNX Runtime, XGBoost, FastAPI, Vertex AI*
- Served **5 models** (XGBoost conversion, KMeans, RandomForest, softmax affinity, TopK recommender) as preloaded **ONNX** graphs behind FastAPI, keeping Python ML off the request path.
- Trained on **10K sessions** and **250 product embeddings**; click-weighted user vectors for **Top-5** recs, delta-synced into BigQuery **VECTOR_SEARCH**.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Go, Java
Agents & Tooling: Model Context Protocol (MCP), multi-agent orchestration, tool calling, Google ADK, LangChain, evals
RAG & Retrieval: RAG, Haystack, Elasticsearch, embeddings, semantic search, SQLite FTS5, BigQuery VECTOR_SEARCH
Full Stack: FastAPI, Next.js, React, React Native, Node.js, SSE, PostgreSQL
ML & Infra: Gemini 2.5 Pro/Flash, GPT-4, ONNX Runtime, PyTorch, Docker, GCP (Vertex AI, Pub/Sub, BigQuery), AWS, pytest, CI/CD